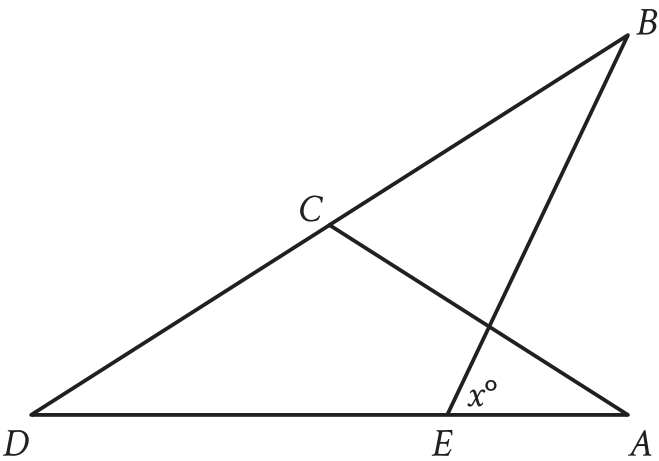


Question ID: 6d99b141

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



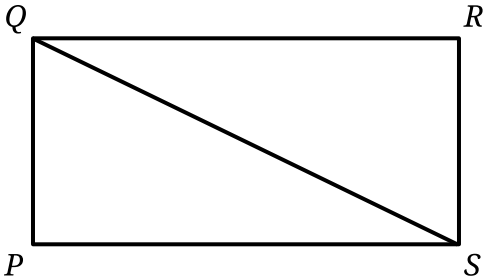
Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is 45° , and the measure of angle ACD is 104° . What is the value of x ?

Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

Question ID: 9912e19f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

Answer

- A. 20°
- B. 45°
- C. 135°
- D. 160°

Question ID: 4b7bb316

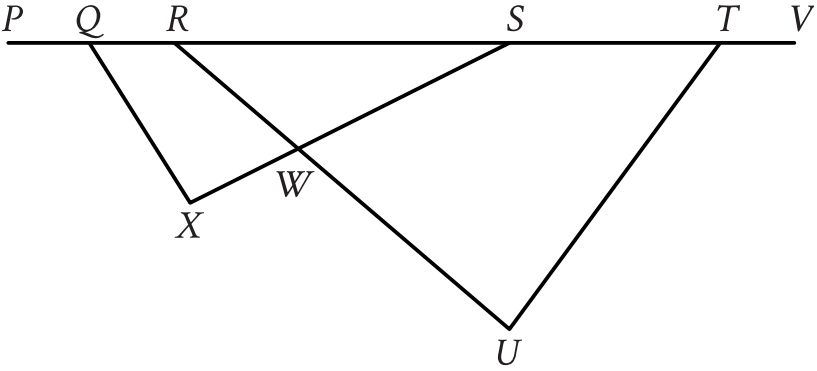
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

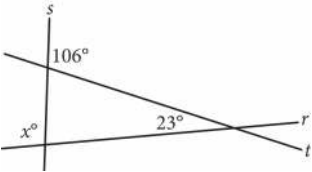
Answer

- A. $\frac{75}{317}$
- B. $\frac{308}{317}$
- C. $\frac{317}{308}$
- D. $\frac{317}{75}$

Question ID: f88f27e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question
Intersecting lines r, s, and t are shown below.



What is the value of x ?

Question ID: e89f63bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

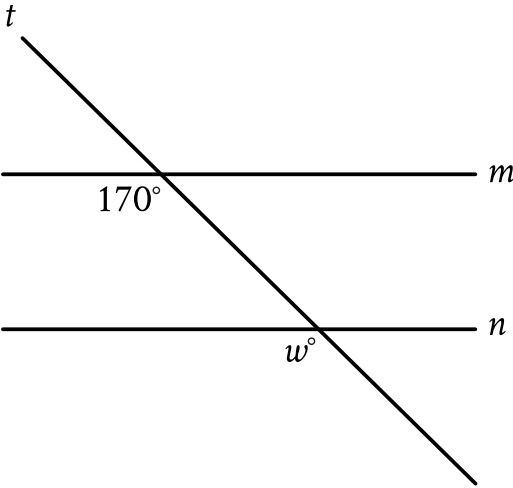
The sum of the measures of two angles in a triangle is **64°**. What is the measure of the third angle?

Answer

- A. **26°**
- B. **52°**
- C. **116°**
- D. **128°**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

Answer

- A. 17
- B. 30
- C. 70
- D. 170

Question ID: f67e4efc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

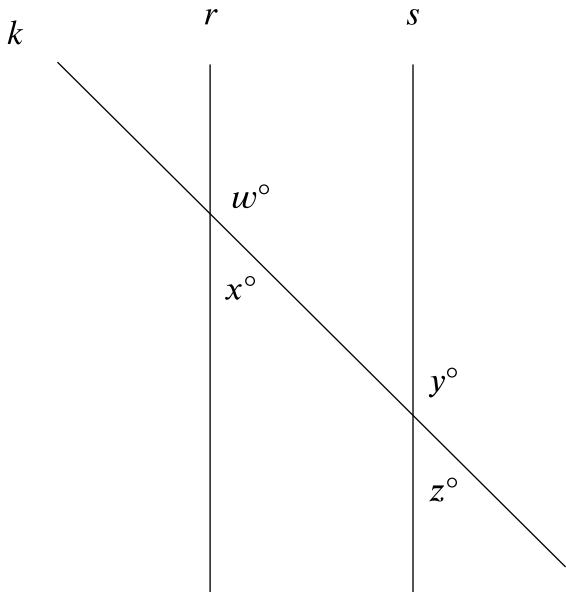
A right circular cylinder has a volume of 45π . If the height of the cylinder is 5, what is the radius of the cylinder?

Answer

- A. 3
- B. 4.5
- C. 9
- D. 40

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line k intersects lines r and s . If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x = 33$
- B. $y = 147$
- C. $w + y = 180$
- D. $y + z = 180$

Choice D is incorrect. The relationship $y + z = 180$ describes a linear pair at line s , which is true regardless of whether lines r and s are parallel.

Question ID: e5c57163

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

Question ID: 33e29881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

Answer

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

Question ID: 93f48423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

What is the area, in square inches, of a rectangle with a length of **7** inches and a width of **6** inches?

Answer

- A. **13**
- B. **20**
- C. **42**
- D. **84**

Question ID: 5252e606

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The side length of a square is **55 centimeters (cm)**. What is the area, **in cm²**, of the square?

Answer

- A. 110
- B. 220
- C. 3,025
- D. 12,100

Question ID: bb560789

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

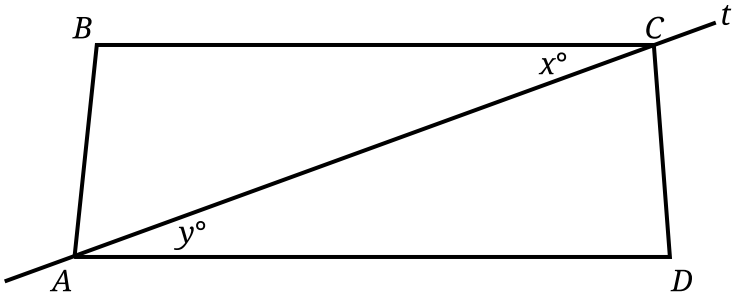
Triangle R has an area of **80 square centimeters (cm^2)**. Square S has side lengths of **4 cm**. What is the total area of triangle R and square S, **in cm^2** ?

Answer

- A. **42**
- B. **44**
- C. **84**
- D. **96**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

Answer

- A. 22
- B. 44
- C. 68
- D. 158

Question ID: 5afbdc8e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

What is the length of one side of a square that has the same area as a circle with radius 2 ?

Answer

- A. 2
- B. $\sqrt{2\pi}$
- C. $2\sqrt{\pi}$
- D. 2π

Question ID: 983412ea

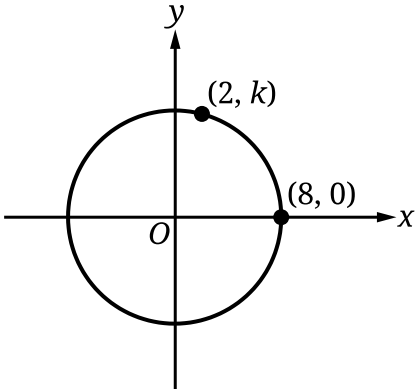
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square prism has a height of **14** units. The volume of the prism is **2,016** cubic units. What is the length, in units, of an edge of the base?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k ?

Answer

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

Question ID: a4c0547f

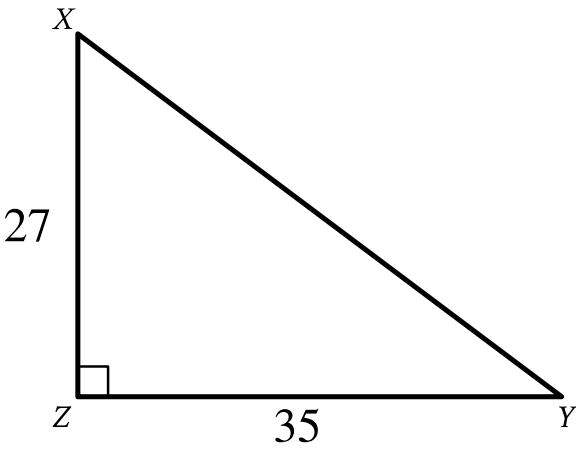
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question
In triangle XYZ , angle Z is a right angle and the length of $\overline{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

- Answer**
- A. 188
 - B. 168
 - C. 84
 - D. 71

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



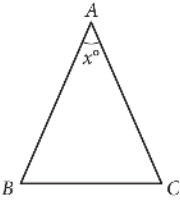
Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

- Answer
- A. $\tan X$
 - B. $\tan Y$
 - C. $\cos X$
 - D. $\cos Y$

Question ID: c8d60e48

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

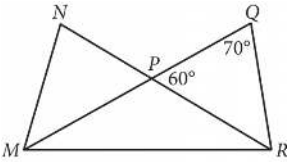
Answer

- A. 36
- B. 46
- C. 58
- D. 70

Question ID: 947a3cde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

Question ID: 1f0b582e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

Answer

- A. **6**
- B. **10**
- C. **14**
- D. **24**

Question ID: deff8a2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

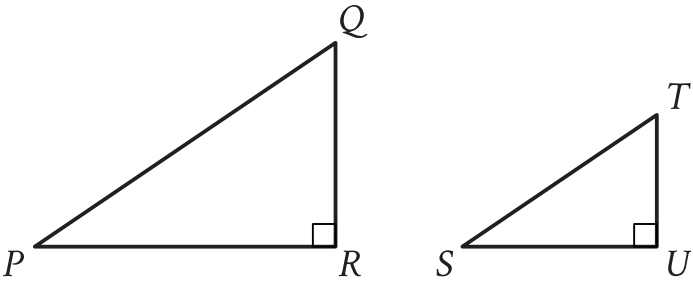
What is the value of n ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



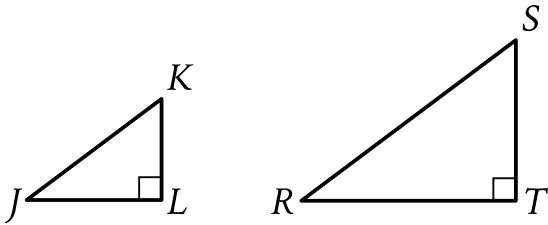
Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

- Answer
- A. 18°
 - B. 72°
 - C. 82°
 - D. 162°

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

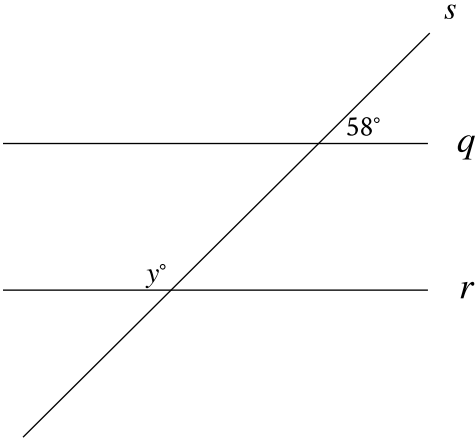
In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\overline{RS}$ is 135. What is the perimeter of triangle RST ?

- Answer
- A. 324
 - B. 540
 - C. 2,916
 - D. 8,100

Question ID: 686b5212

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

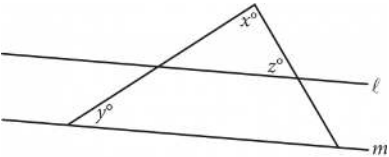


Note: Figure not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

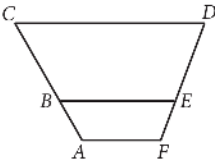
Answer

- A. 120
- B. 100
- C. 90
- D. 80

Question ID: 81b664bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

Answer

- A. 16.8
- B. 17.5
- C. 18.4
- D. 19.6

Question ID: 59cb654c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

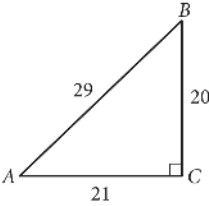
The area of a square is ~~64~~ square inches. What is the side length, in inches, of this square?

Answer

- A. ~~8~~
- B. ~~16~~
- C. ~~64~~
- D. ~~128~~

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



In the figure above, what is the value of $\tan(A)$?

Answer

- A. $\frac{20}{29}$
- B. $\frac{21}{29}$
- C. $\frac{20}{21}$
- D. $\frac{21}{20}$

Question ID: e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A circle has a radius of ~~4~~3 meters. What is the area, in square meters, of the circle?

Answer

- A. $\frac{43\pi}{2}$
- B. ~~4~~3 π
- C. 86 π
- D. 1,849 π

Question ID: 96467fea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question
Circle N has a radius of 7 millimeters (**mm**). Circle M has an area of 64π **mm**². What is the total area, in **mm**², of circles N and M?

- Answer**
- A. 113π
- B. 92π
- C. 78π
- D. 15π

Question ID: 858fd1cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(5, -4)$. Which of the following points also lies on line t ?

Answer

- A. $(0, \frac{6}{5})$
- B. $(4, 7)$
- C. $(10, 2)$
- D. $(11, 1)$

Question ID: 9adb86ed

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question
Points Q and R lie on a circle with center P . The radius of this circle is **9** inches. Triangle PQR has a perimeter of **31** inches. What is the length, in inches, of $\overline{QR}$?

- Answer**
- A. $13\sqrt{2}$
 - B. 13
 - C. $9\sqrt{2}$
 - D. 9

Question ID: 1dc7e423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

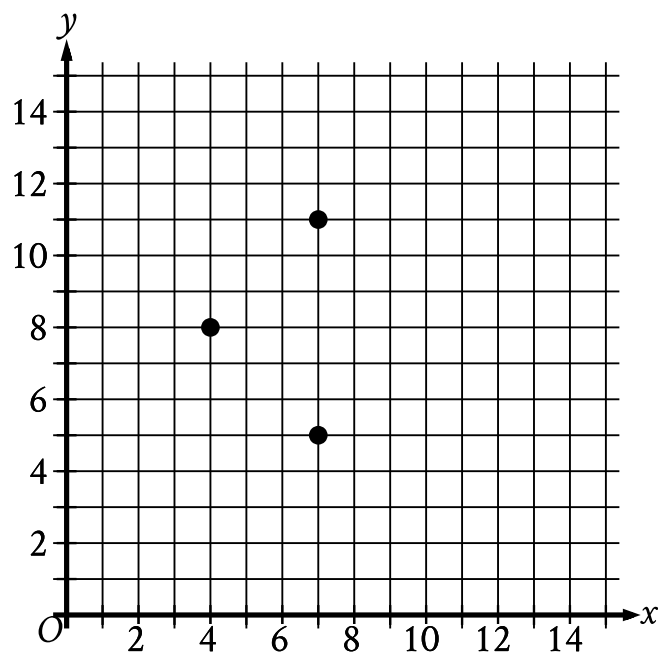
In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

Answer
A. 3

B. 6

C. 7

D. 9

Choice D is incorrect. This is the value of k for which $k\pi$ represents the area of the circle, in square units, not the circumference of the circle, in units.

Question ID: ec5d4823

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

Question ID: 9966235e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has an edge length of **68** inches. A solid sphere with a radius of **34** inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

Answer

- A. 149,796
- B. 164,500
- C. 190,955
- D. 310,800

Question ID: 2b41a4c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. What is the volume, in cubic meters, of the prism?

Question ID: a0369739

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

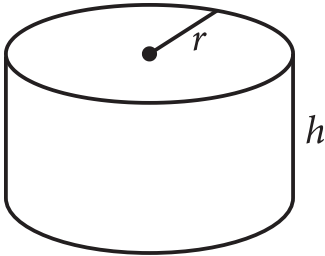
In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

Answer

- A. $\frac{15}{38}$
- B. $\frac{15}{23}$
- C. $\frac{23}{15}$
- D. $\frac{38}{15}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is **392** times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

Answer

- A. $R = 8r$ and $H = 7h$
- B. $R = 8r$ and $H = 49h$
- C. $R = 7r$ and $H = 8h$
- D. $R = 49r$ and $H = 8h$

Question ID: cbe8ca31

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

Answer

- A. 58°
- B. 74°
- C. 122°
- D. 212°

Question ID: 3b931fb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

Question ID: 94364a79

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

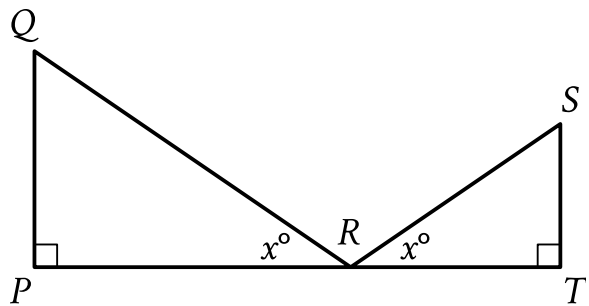
Two nearby trees are perpendicular to the ground, which is flat. One of these trees is **10** feet tall and has a shadow that is **5** feet long. At the same time, the shadow of the other tree is **2** feet long. How tall, in feet, is the other tree?

Answer

- A. **3**
- B. **4**
- C. **8**
- D. **27**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are **14**, **15**, **20**, and **25**, respectively. What is the length of $\overline{SR}$?

Answer

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

$(x - 3)^2 + (y + 16)^2 = 289$

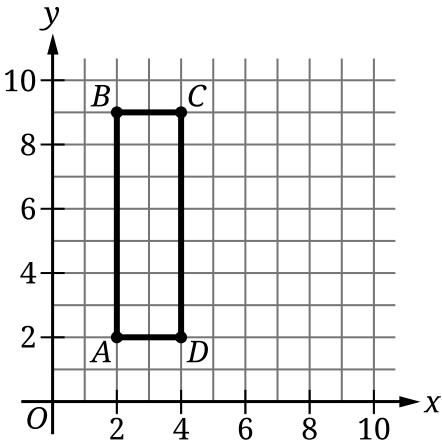
The graph of the given equation is a circle in the xy -plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q ?

Answer

- A. -17
- B. -3
- C. 3
- D. 17

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



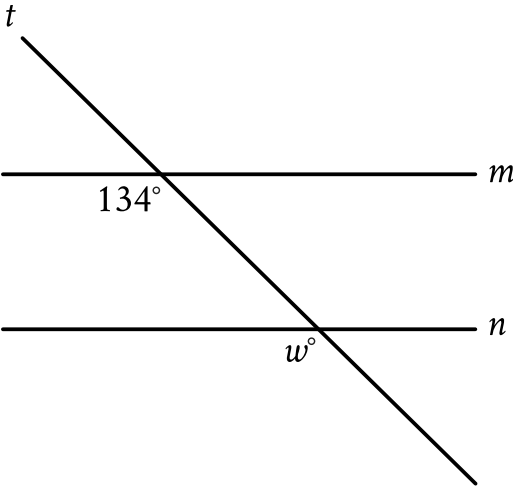
Rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. What is the area, in square units, of rectangle $ABCD$?

Answer

- A. **5**
- B. **9**
- C. **14**
- D. **28**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

Answer

- A. 13
- B. 34
- C. 66
- D. 134

Question ID: 055aafe7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

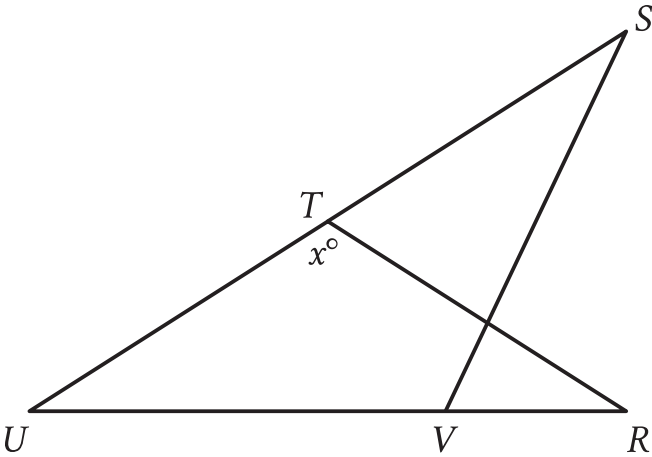
Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

Answer

- A. 204
- B. 182
- C. 60
- D. 12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

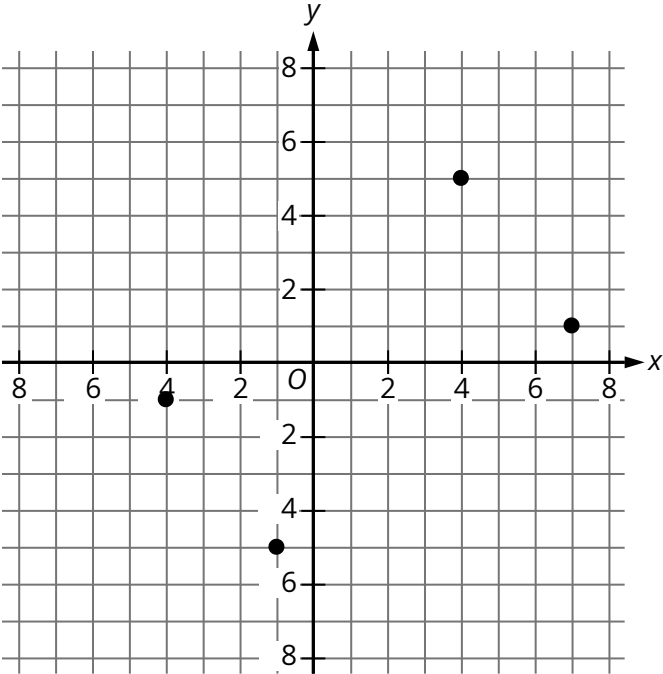


Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

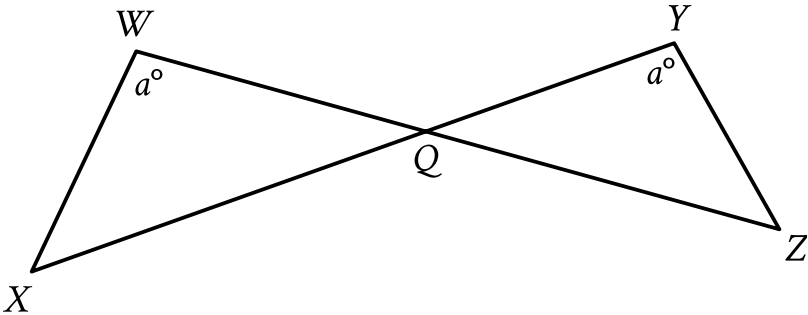
Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

Question ID: 8493fd10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

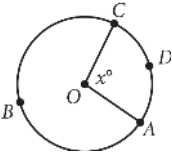
Answer

- A. 73
- B. 71
- C. 37
- D. 35

Question ID: c8345903

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

Answer

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

Question ID: 901c3215

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

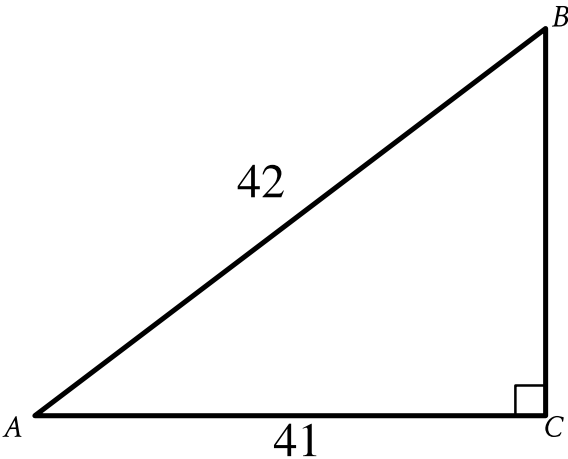
In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

Answer

- A. The measure of angle A
- B. The length of side AB
- C. The lengths of sides BC and EF
- D. No additional information is necessary.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

Answer

- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

Question ID: 85f1892d

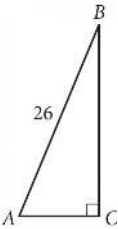
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle XYZ , angle Y is a right angle, the measure of angle Z is 33° , and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant, what is the value of k ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Question ID: f7dbde16

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

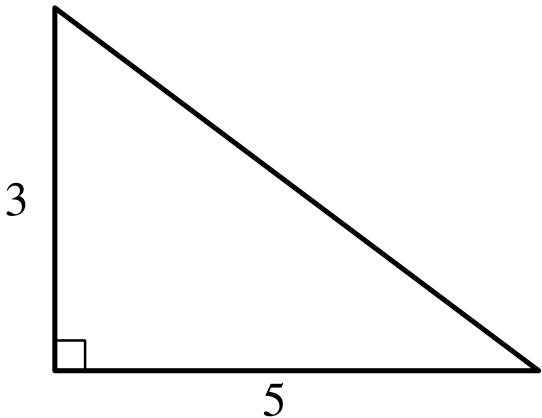
In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

Answer

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question



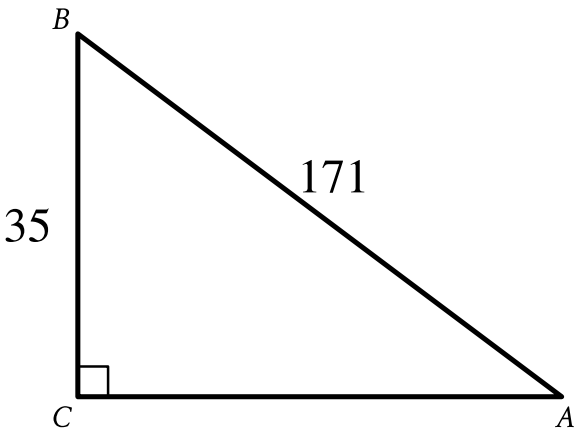
Note: Figure not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

Question ID: 87a9a2d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

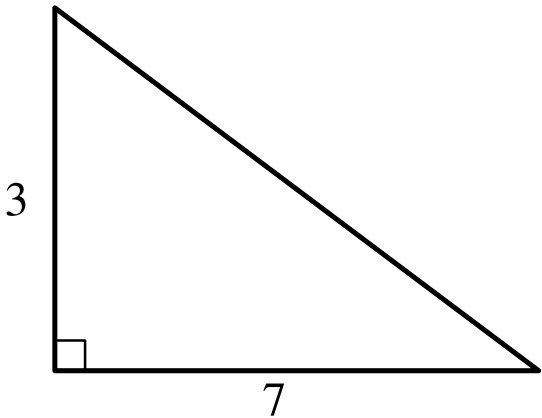
Answer

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

Question ID: e6f2ace7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



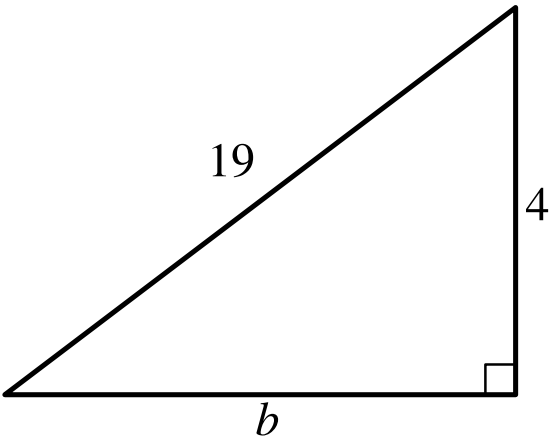
Note: Figure not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

- Answer**
- A. 3.2
 - B. 5
 - C. 7.6
 - D. 20

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

Answer
A. $4b = 19$

B. $4 + b = 19$

C. $4^2 + b^2 = 19^2$

D. $4^2 - b^2 = 19^2$

Question ID: 76c73dbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

Question ID: 58c26db8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

Answer

- A. 9
- B. $9\sqrt{2}$
- C. 18
- D. $18\sqrt{2}$

Question ID: e336a1d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

Answer

- A. **164**
- B. 1,681
- C. 10,086
- D. 68,921

Question ID: e4b4e9ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of the edge of the base of a right square prism is **6** units. The volume of the prism is **2,880** cubic units. What is the height, in units, of the prism?

Answer

- A. $4\sqrt{30}$
- B. **36**
- C. $24\sqrt{5}$
- D. 80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

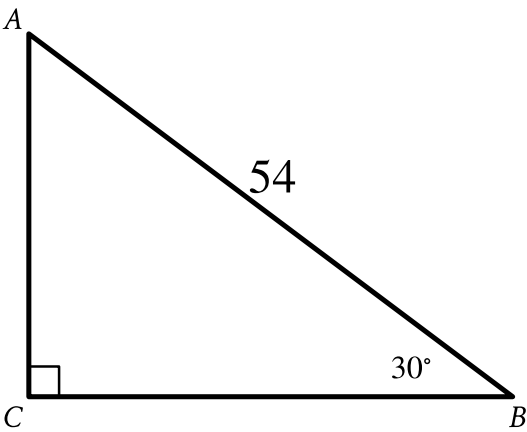
A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder?

Answer

- A. 16π
- B. 96π
- C. 192π
- D. 768π

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

Answer

- A. $\frac{\sqrt{3}}{54}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $27\sqrt{3}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

Answer

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

Question ID: 151eda3c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

Answer

- A. 16,000 π
- B. 20,000 π
- C. 25,000 π
- D. 31,250 π

Question ID: 35d37640

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

Answer

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

$(x + 4)^2 + (y - 19)^2 = 121$

The graph of the given equation is a circle in the xy -plane. The point (a, b) lies on the circle. Which of the following is a possible value for a ?

Answer

- A. -16
- B. -14
- C. 11
- D. 19

Question ID: 167aff9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is **58 square centimeters (cm^2)**, and the surface area of right rectangular prism Y is **1,450 cm^2** . The volume of right rectangular prism Y is **1,250 cubic centimeters (cm^3)**. What is the sum of the volumes, **in cm^3** , of right rectangular prism X and right rectangular prism Y?

Question ID: 2266984b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$x^2 + 20x + y^2 + 16y = -20$

The equation above defines a circle in the xy-plane. What are the coordinates of the center of the circle?

Answer

- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

Question ID: d0b6d927

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A rectangle has an area of **63** square meters and a length of **9** meters. What is the width, in meters, of the rectangle?

Answer

- A. **7**
- B. **54**
- C. **81**
- D. **567**

Question ID: d621cffb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

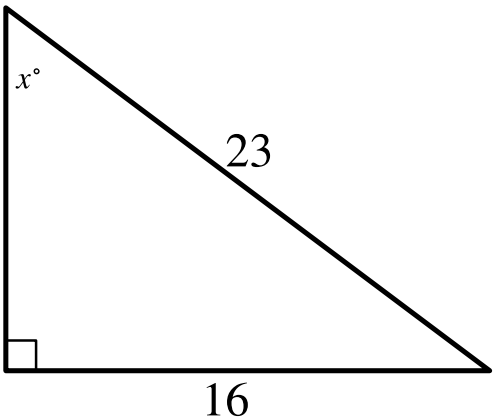
Answer

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

Question ID: 1429dcd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



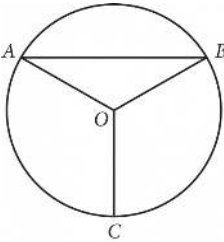
Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

Question ID: 69b0d79d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

Answer

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Question ID: 5a7e3b46

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is **136** millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

Answer

- A. **34**
- B. **102**
- C. **136**
- D. **170**

Question ID: ebbf23ae

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

Question ID: d7a8aa9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

Answer

- A. $\frac{29}{28}$
- B. 1
- C. $\frac{28}{29}$
- D. 0

Question ID: a2659088

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a height of **8 meters (m)** and a base with a radius of **12 m**. What is the volume, **in m³**, of the cylinder?

Answer

- A. **8π**
- B. **20π**
- C. **768π**
- D. **$1,152\pi$**

Question ID: 502d9690

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

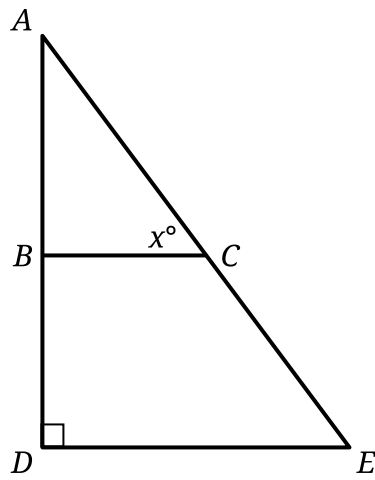
Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is **648** square inches, and the area of rectangle $EFGH$ is **72** square inches. The length of the longest side of rectangle $ABCD$ is **36** inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

Answer

- A. **4**
- B. **9**
- C. **12**
- D. **36**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 147 and the length of $\overline{AE}$ is 245, what is the value of $\tan x^\circ$?

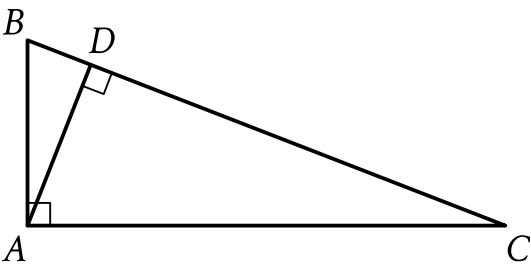
Answer

- A. $\frac{147}{196}$
- B. $\frac{147}{245}$
- C. $\frac{196}{147}$
- D. $\frac{245}{147}$

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

Question ID: 6b4707aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

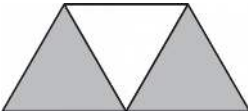
Question

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

Question ID: 4c95c7d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

Answer

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Question ID: b8a225ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

Question ID: b0a72bdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

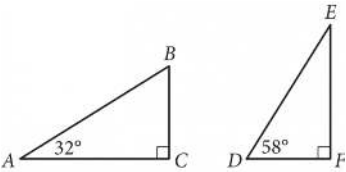
What is the diameter of the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

Answer

- A. 4
- B. 8
- C. 16
- D. 32

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

Answer

- A. $\frac{DE}{DF}$
- B. $\frac{DF}{DE}$
- C. $\frac{DF}{EF}$
- D. $\frac{EF}{DE}$

Choice A is incorrect because $\frac{DE}{DF}$ is the reciprocal of the ratio $\frac{BC}{AB}$. Choice C is incorrect because $\frac{DF}{EF} = \frac{BC}{AC}$, not $\frac{BC}{AB}$. Choice D is incorrect because $\frac{EF}{DE} = \frac{AC}{AB}$, not $\frac{BC}{AB}$.

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

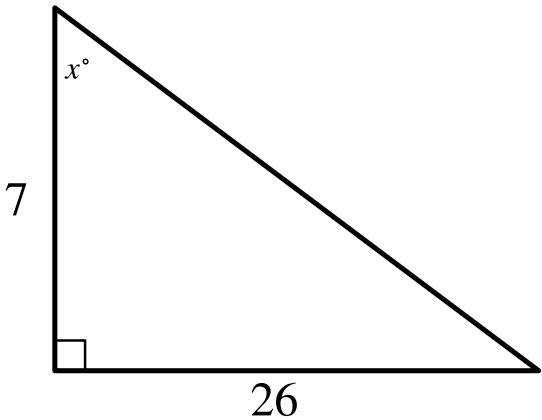
Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Question ID: 64c1f044

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

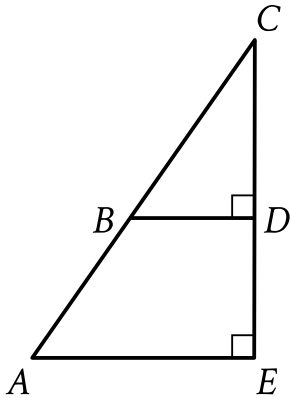
In the triangle shown, what is the value of $\tan x^\circ$?

Answer

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26(BD)$. What is the measure of angle CAE ?

Answer

- A. $(26 \cdot 57)^\circ$
- B. $(26 + 57)^\circ$
- C. 57°
- D. 26°

Question ID: c447367f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

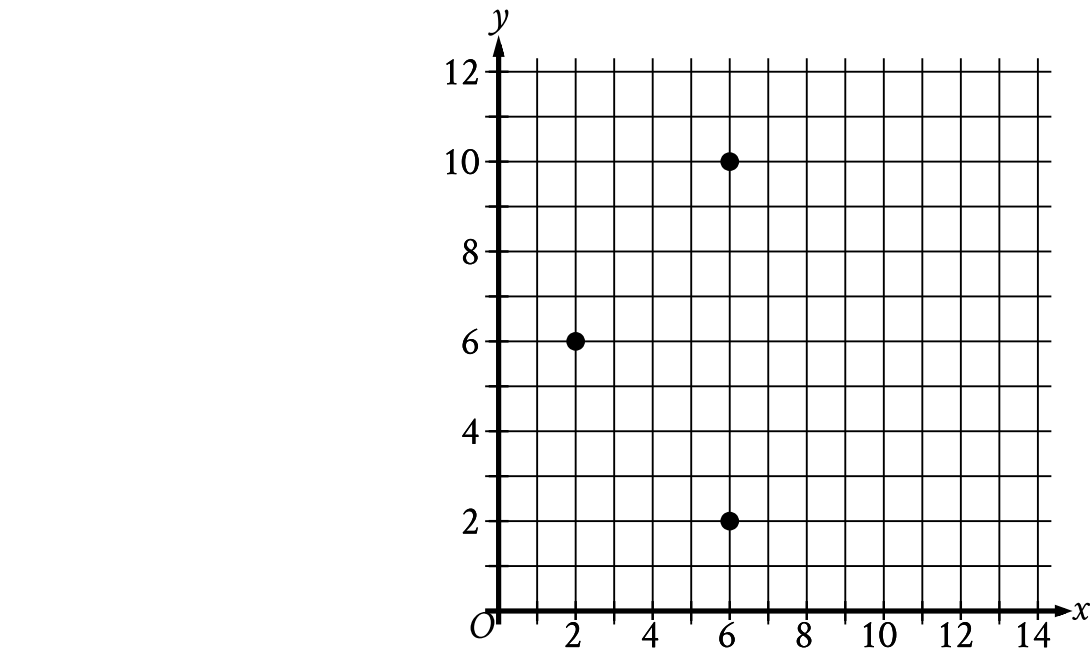
A circle in the xy -plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

Answer

- A. $(x - 19)^2 + (y - 18)^2 = 81k$
- B. $(x - 19)^2 + (y - 18)^2 = 81k^2$
- C. $(x - 19)^2 + (y - 18)^2 = 9k$
- D. $(x - 19)^2 + (y - 18)^2 = 9k^2$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

Question ID: 9fec9d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The floor of a ballroom has an area of **600** square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

Answer

- A. **6**
- B. **10**
- C. **60**
- D. **150**

Question ID: ba8ca563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

Question ID: 5011b039

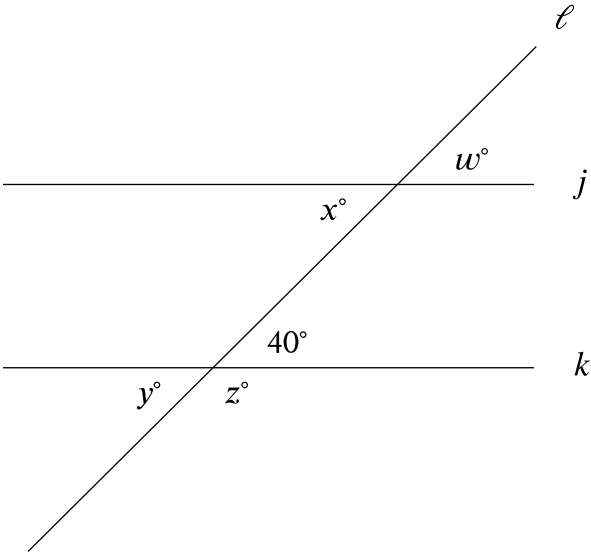
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

What is the radius of the circle in the xy -plane defined by $(x + 2)^2 + (y + 5)^2 = 169$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- Answer
- A. $w = 40$
 - B. $x = 140$
 - C. $y = 40$
 - D. $z = 140$

Question ID: b1e1c2f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In right triangle ABC , angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE ?

Question ID: e3158182

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

Answer

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

Question ID: 267f82b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

A circle in the xy -plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

Answer

- A. The center is at $(19, 17)$, and the radius is 8 .
- B. The center is at $(19, 17)$, and the radius is $8k$.
- C. The center is at $(19, 17)$, and the radius is 64 .
- D. The center is at $(19, 17)$, and the radius is $64k$.

Question ID: 0d43db90

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

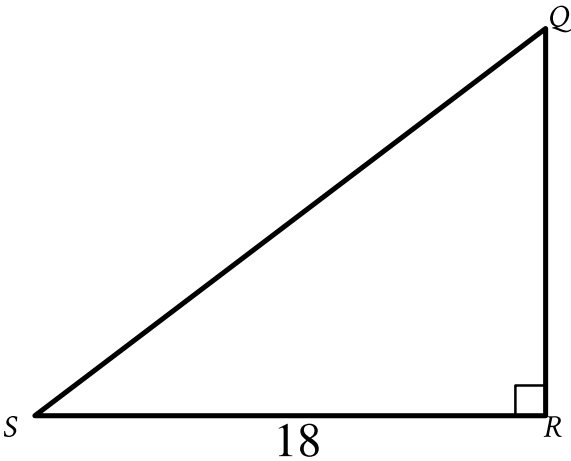
The perimeter of triangle ABC is **17** inches, the length of side AB is **4** inches, and the length of side AC is **7** inches. What is the length, in inches, of side BC ?

Answer

- A. **4**
- B. **6**
- C. **7**
- D. **11**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $18 \cos Q$
 - B. $18 \sin Q$
 - C. $\frac{18}{\cos Q}$
 - D. $\frac{18}{\sin Q}$

Question ID: e0874bc2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of $\overline{XY}$?

Answer

- A. 2
- B. 18
- C. 55
- D. 162

Question ID: a4bd60a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The perimeter of an equilateral triangle is ~~624~~ centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

Question ID: 899c6042

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

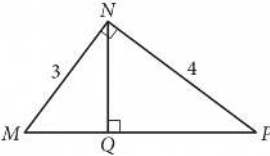
Answer

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$

Question ID: 740bf79f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, what is the length of $\overline{NQ}$?

Answer

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

Question ID: 0e40dfb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A rectangle has a length of **3** units and a width of **39** units. Which expression gives the area, in square units, of this rectangle?

Answer

- A. **2(3 + 39)**
- B. **2(3 · 39)**
- C. **3 + 39**
- D. **3 · 39**

Question ID: 3b225698

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

Answer

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Question ID: 249d3f80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle ROS ?

Question ID: fc5ef8d3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

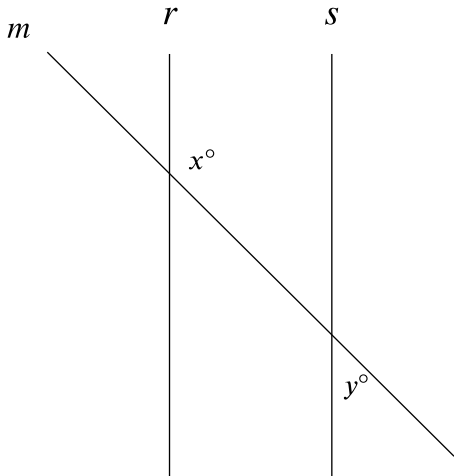
What is the length of $\overline{XY}$?

Answer

- A. 2
- B. 6
- C. 18
- D. 56

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- Answer
- A. $x < 115$
 - B. $x > 115$
 - C. $x + y < 180$
 - D. $x + y > 180$

Question ID: 38517165

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?